

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/MICROBIOLOGY)

SEMESTER II EXAMINATION (REGULAR)

BT708: Industrial Biotechnology

Date: 06/05/2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

Provide all necessary information as required in the answer sheet.

1. Attempt all questions.
2. Questions in **Part A** should be answered in the question paper itself. Please do **not** write your candidate ID number on the question paper for Part A.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.
4. Use of cell phones is strictly prohibited.
5. Use of non-programmable calculators may be allowed if needed.

PART A

Total Marks:20

Choose the correct option and put $\sqrt{\quad}$ mark in front of it:

Where the target details are initially less important and the activity is focused on screening large number of chemical molecules with the hope of finding a winner is _____ approach

(a) Top down (b) bottom up (c) Top up (d) bottom down

2. 98 % of the biotechnology products are related to _____
(a) health care (b) Transgenic plants (c) cosmetic (d) a, b and c
3. IND stands for _____
(a) Investigational new design (b) Investigational new drug (c) Investment in new drug
(d) a, b and c
4. _____ is the stage between discovery and commercialization that covers testing and optimization, combined with manufacture of the product to high quality in pharmaceutical company.
(a) drug development (b) clinical trials (c) non clinical trials (d) a, b and c
5. Citrate is served as inhibitor of following enzyme
(a) Hexokinase (b) phosphoglucose isomerase (c) phosphofructokinase (d) a and b both
6. Ethanol and water form azeotrope at following molar percent of ethanol
(a) 88 (b) 98 (c) 80.9 (d) none of the above
7. Which of the following enzyme is disappear when ferrocyanide is added in citric acid production ?
(a) aconitate hydratase (b) isocitrate dehydrogenase (c) isocitric lyase (d) all the three
8. Following amino acid does not belongs to aspartate family
(a) Lysin (b) Methionine (c) Threonine (d) glutamine
9. A – 10 auxotrophic mutant used in rifamycin production lack
(a) transketolase (b) shikimate phosphorylation (c) both (d) none of the above

10. Amylase belong to the class
(a) Lyases (b) Hydrolases (c) Transferases (d) Oxidoreductases
 11. Xanthan gum contain repeated unit of primary structure
(a) 1 glucose, 1 mannose and 1 glucuronic acid (b) 2 glucose, 1 mannose and 2 glucuronic acid
(c) 1 glucose, 2 mannose and 2 glucuronic acid (d) 2 glucose, 2 mannose and 1 glucuronic acid
 12. Siderophores chelate the following ion
(a) iron (b) Manganese (c) potassium (d) copper
 13. Drug used in the treatment of disease is known as _____
(a) pharmaceutical (b) Nutraceuticals (c) botanical (d) none of a, b and c
 14. Ginseng is an example of _____ Nutraceutical
(a) Dietary supplement (b) herbal (c) Nutrient (d) none of a, b and c
 15. Copy right is an IPR used for
(a) arts (b) literature (c) movies (d) a, b, and c
- State True or False
16. Genentech licensed the human insulin technology to Eli Lilly, where it was named Humulin. TRUE/FALSE.
 17. As part of their routine inspection of manufacturing facilities, regulatory personnel will usually inspect a sample of these records, to ensure conformance to GMP. TRUE/FALSE.
 18. Validation carried out by review of historical records is concurrent validation. TRUE/FALSE.
 19. Good manufacturing practice is a part of quality control. TRUE/ FLASE
 20. Scientific advisers advise the industrial persons in Scientific matters. TRUE/ FLASE

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PART B

Total Marks: 50

Section I

- Q 1 A Answer any one [4]
(i) Types of investors
(ii) Difference between academic and commercial science
B Answer any one [6]
(i) Types of government support given to biotechnology company
(ii) Note on Clinical trials
- Q 2 A Define : (i) Synbiotic (ii) Probiotic (iii) Recombinant protein [5]
(iv) Herbal (v) Pharmaceutical
OR
Define Nutraceuticals and write note on types of Nutraceuticals.
B Write Note on any one [5]
(i) Probiotics
(ii) Recombinant proteins

Section II

- | | | | |
|-----|---|--|-----|
| Q 3 | A | Answer in detail. (Any TWO)
(i) Write down the fermentation kinetic of ethanol production
(ii) Lysine production with threonine auxotrophs
(iii) Production of Vitamin B12 by <i>Propionibacterium shermani</i> | [6] |
| | B | Answer the following (Any TWO)
(i) Crabtree effect
(ii) Role of biotin in L-glutamic acid production
(iii) Write down azeotropic alcohol production using vacuum rectification | [4] |
| Q 4 | A | Note on following (any two)
(i) Write down the recovery of thurigiensis and mention the role of lactose and acetone
(ii) Xanthan recovery
(iii) Nodule formation | [6] |

B Briefly explain (any two)

[4]

- (i) Role of Barbiturate in rifamycin fermentation
- (ii) Write down the mechanism of iron transport
- (iii) Rifamycin B

Q 5 A Block diagram of Manufacturing facility of biotechnology based product.

[5]

OR

Define validation and explain the types of validation

B Answer any one

[5]

- (i) Mention the main areas of GMP and explain any two in details.
- (ii) Note on Good Laboratory Practice